Notes on Laplacian-Coupled Networks of NDR Memristive FitzHugh–Nagumo Motifs

Contents

1 What these notes are about

These are working notes on a concrete physical question: what happens when you take N active memristive neuron circuits and wire their exposed voltage nodes together through an ordinary resistor network? The circuits are the NDR memristive FitzHugh–Nagumo motifs of Pei et al. [7]. One reason this question matters is that it has become clear in our work that a single memristor, one with a unique fixed point, does not spontaneously develop multiple fixed points when you couple it to others of the same kind — roughly, $1^M = 1$. So if you want associative memory, you need the building blocks to already be bistable on their own, before coupling enters the picture. That is why we focus on NDR motifs: each one is natively bistable and acts as a one-bit memory element. The coupling then has a chance to do something interesting.

The short physical answer is that the resistor network acts as a diffusive coupling, with a strength set by the ratio of the inter-node conductance to the nodal capacitance. The longer answer, which these notes work out carefully, is that any stable resting state of an isolated motif survives that coupling and stays stable, as long as the coupling is not too strong. The argument runs from the device level upward: we characterise the NDR memristor, assemble the FitzHugh–Nagumo motif from it, and apply Kirchhoff's laws to get the equations of motion for the whole network. The result is a smooth perturbation of N non-interacting copies of the same two-dimensional oscillator. Two standard tools — the Implicit Function Theorem and the continuity of eigenvalues — then do the heavy lifting.

The same framework also covers the case where small voltage sources sit in series with the coupling resistors. Circuit-wise, these model synaptic reversal potentials; in the dimensionless FN equations they appear as a node-wise bias that shifts the fixed points without touching their stability.

The logical skeleton

It is worth spelling out the structure of the argument before diving into circuit algebra, because otherwise it is easy to get lost in the bookkeeping.

When the coupling resistors are absent — or equivalently, when their resistance is infinite — the N motifs do not interact. Each motif i obeys its own two-dimensional ODE,

$$\dot{X}_i = G(X_i), \qquad X_i = (x_i, y_i)^T \in \mathbb{R}^2, \tag{1}$$

and the joint state $X = (X_1, \ldots, X_N)$ is just the direct product $\dot{X} = F_0(X) := (G(X_1), \ldots, G(X_N))$.

Before saying anything about stability, though, we need to be clear about which dynamical regime we are in, because the FitzHugh–Nagumo system has two qualitatively different behaviours and only one of them is relevant here.

In the *oscillatory regime*, the single fixed point sits on the middle, unstable branch of the cubic voltage nullcline, where the differential conductance is positive and the Jacobian has a positive trace. The fixed point is an unstable spiral, and a stable limit cycle lives around it. In plain language, the motif oscillates indefinitely and never settles anywhere. There is nothing to perturb stably, so the programme of these notes simply does not apply.

In the *bistable regime* — the one we stay in throughout — the linear recovery nullcline cuts the cubic voltage nullcline at three distinct points. The two outer intersections, p_1 on the left branch and p_3 on the right, lie where $|x_j| > 1$, in the NDR region where the differential conductance is negative. There, by the Routh–Hurwitz criterion, both p_1 and p_3 are linearly stable fixed points; small perturbations decay exponentially. The middle intersection p_2, with $|x_2| < 1$, is always a saddle: it is unstable and acts only as the boundary between the two basins of attraction. The motif does not oscillate in this regime — the limit cycle was destroyed as the parameters moved into the bistable window.

The two regimes are separated by bifurcations in the (a, b, z) parameter space. As the stimulus z increases past a saddle-node value, the outer fixed points collide and annihilate, the bistable window closes, and the system transitions to monostable oscillation. The Pei et al. device is designed to operate in the bistable regime, where it acts as a two-state memory element rather than a clock.

A fixed point p_j is *Hurwitz* if all eigenvalues of the Jacobian $DG(p_j)$ have strictly negative real part. In the bistable regime, p_1 and p_3 satisfy this. The product system therefore has $M^N = 3^N$ fixed points in total, of which 2^N are Hurwitz — those in which every node sits at p_1 or p_3 — and the remaining $3^N - 2^N$ involve at least one saddle and are unstable.

Now we turn on the resistors. We introduce a single overall conductance scale $\bar{g} = 1/R_0$ and write the conductance Laplacian as $L_G = \bar{g}L$, where L is a dimensionless graph Laplacian with entries of order one. The coupling parameter is

$$\varepsilon = \frac{\bar{g}}{C_0} = \frac{1}{R_0 C_0}, \tag{2}$$

the inverse of the RC time constant of the intermediate network relative to the nodal capacitance. Weak coupling means $\varepsilon \ll 1$, that is, $R_0 \gg 1/C_0$. Sending $R_0 \to \infty$ recovers the uncoupled system. The coupled network then takes the form

$$\dot{X} = F_0(X) + \varepsilon H(X), \tag{3}$$

a smooth $O(\varepsilon)$ deformation of the product system.

The key property of each Hurwitz fixed point p_j is that $\det DG(p_j) \neq 0$. This means $\det DF_0(P_\alpha) \neq 0$ for every product fixed point P_α, and the Implicit Function Theorem guarantees that, for small enough ε, each P_α deforms continuously into a nearby fixed point $P_\alpha(\varepsilon)$ of the coupled system. Eigenvalue continuity then ensures that, since all eigenvalues of $DF_0(P_\alpha)$ started in the open left half-plane with some spectral gap $\gamma_\alpha > 0$, they cannot cross the imaginary axis for small ε. Taking the minimum over the finite set of multi-indices gives a uniform threshold $\varepsilon_* > 0$ valid simultaneously for all 2^N Hurwitz product fixed points.

The upshot: *work in the bistable regime, where each isolated motif has two stable fixed points and no limit cycle; turn on weak coupling; all 2^N Hurwitz fixed points survive and remain stable*, as long as $\varepsilon = 1/(R_0 C_0)$ is small enough.

2 The NDR memristor and the FitzHugh–Nagumo motif

2.1 The NDR memristor

An active negative-differential-resistance (NDR) device is a two-terminal element that, over some voltage range, delivers more current as the voltage is *reduced*. This is obviously impossible for an ordinary resistor and requires either an internal energy source or a quantum-mechanical transport mechanism. Writing the terminal current as $I = \phi(V, \xi)$, where V is the applied voltage and ξ is a vector of internal state variables, the NDR region is where

$$g_{\text{diff}}(V, \xi) = \frac{\partial \phi}{\partial V}(V, \xi) < 0. \tag{4}$$

The device can then supply incremental power to the surrounding circuit, which is what makes NDR elements useful for sustaining oscillations: a negative resistance in a loop with a capacitor compensates dissipation and allows self-amplification. This is exactly what happens in voltage-gated sodium channels, where the conductance increases with depolarisation — a positive feedback whose mathematical fingerprint is a folded-cubic I–V curve.

The same element is called *memristive* when the internal state has its own dynamics,

$$\dot{\xi} = q(V, \xi), \tag{5}$$

so the current at any instant depends on the past history of the voltage through ξ [9]. Together, (4) and (5) define the class of devices we work with.

Definition 1 (NDR memristive element)**.** *A two-terminal element is an* NDR memristive element *if its terminal current is* $I = \phi(V, \xi)$ *with* $\partial_V \phi < 0$ *on an open voltage interval, and its internal state obeys* $\dot{\xi} = q(V, \xi)$*.*

The device used throughout these notes is the NDR memristor of Pei et al. [7], built from an $AlAs/In_{0.8}Ga_{0.2}As/AlAs$ quantum-well heterostructure on an InP substrate. The $In_{0.8}Ga_{0.2}As$ well is about 3 nm thick, thin enough that electrons are confined into discrete transverse energy levels. When the applied bias aligns the Fermi level of the emitter with one of these levels, resonant tunneling through the double AlAs barrier produces a sharp current peak; increasing the bias detunes the resonance and the current falls — this is the NDR region. Atomic vacancies in the well trap electrons on the upswing and release them on the downswing, producing the box-type hysteretic I–V loop that is the signature of memristive behaviour.

Because the switching is governed by band structure rather than filament formation or ion migration, the device is unusually robust: cycle-to-cycle variation in peak and valley currents is below 0.3%, endurance exceeds 10^{11} cycles at room temperature and 10^9 cycles at 400 °C, and the peak-to-valley current ratio remains above 1.86 across that entire range [7]. In our language, ξ encodes the occupancy of the vacancy trapping sites, and $\dot{\xi} = q(V, \xi)$ is slow compared with the oscillation period, which is why the device shows hysteresis rather than instantaneous switching. The effective negative resistance in the NDR region is

$$R_n = \frac{2\Delta V}{3\Delta I}, \qquad \Delta V = V_p - V_v, \quad \Delta I = I_p - I_v, \tag{6}$$

giving $\partial_V \phi \approx -1/|R_n| < 0$ in the NDR region, as required by (4).

2.2 The FitzHugh–Nagumo circuit motif

The biological inspiration is the Hodgkin–Huxley picture of the action potential [3]: fast inward Na^+ current drives depolarisation, slower outward K^+ current drives repolarisation, and the interplay between them creates the spike. In the Pei et al. circuit [7], two NDR memristors D_1 and D_2 with different threshold voltages $V_{\text{th},1} < V_{\text{th},2}$ are connected in parallel and placed in series with an inductor L_1. The device D_1 with the lower threshold plays the fast Na^+-like role; D_2 switches later and provides the slower K^+-like repolarising current. The inductor acts as the inductive memory element that, together with the negative resistance, sustains the oscillation. Importantly, the intrinsic quantum-well capacitance of each device takes the place of the membrane capacitance, so no external capacitor is needed.

The aggregate I–V characteristic of the $D_1 \| D_2 + L_1$ combination is $I_{\text{motif}} = \phi(V, \xi)$, with $\xi = (\xi_1, \xi_2)$ encoding both NDR element states. The slow K^+-like state ξ_2 becomes the recovery variable y in the FitzHugh–Nagumo equations — we make that identification explicit in Section 4. The motif has been shown to reproduce nine experimentally verified FN neuron behaviours, including phasic spiking, tonic bursting, spike accommodation, anodal break excitation, and all-or-nothing firing [7].

3 Circuit equations for the coupled network

3.1 Network topology and the incidence matrix

We consider N identical NDR motifs whose exposed nodes are connected through a passive resistive network (see Fig. 1).

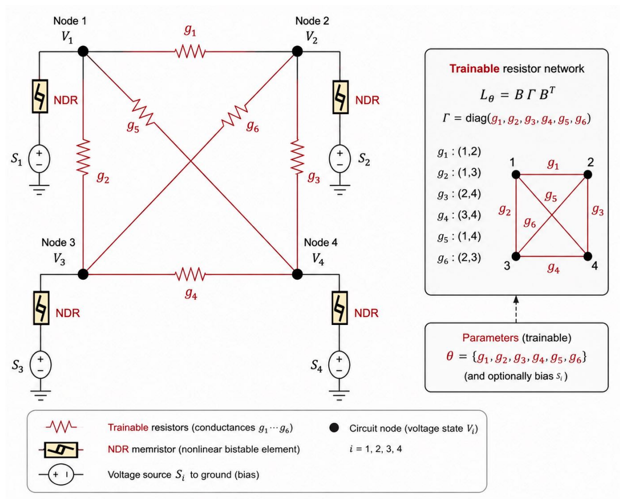

Figure 1: Schematic of the coupled NDR network. Each node i carries an NDR motif M_i and a local bias source s_i; the exposed nodes are linked by the intermediate resistive network with resistors $R_{12} = g_1^{-1}$, $R_{23} = g_6^{-1}$, etc.

Each motif i exposes one terminal node at voltage v_i and has its other terminal grounded through the NDR branch, which also contains a fixed series bias source s_i. This bias shifts the effective voltage seen by the NDR element and will later map onto the applied stimulus z_i in the FitzHugh–Nagumo equations; it can also encode a reversal potential or a dc offset.

The intermediate network has E edges. We choose an arbitrary orientation for each edge and encode the topology in the directed incidence matrix

$$B \in \mathbb{R}^{N \times E}, \qquad B_{ie} = \begin{cases} +1, & \text{edge } e \text{ enters node } i, \\ -1, & \text{edge } e \text{ leaves node } i, \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

The choice of orientation is arbitrary; the physics does not depend on it. We collect node voltages, edge currents, and conductances:

$$v = (v_1, \ldots, v_N)^T, \quad j = (j_1, \ldots, j_E)^T, \quad G = \mathrm{diag}(g_1, \ldots, g_E). \tag{8}$$

For motif i, let i_i be the NDR branch current oriented from the exposed node toward ground. The voltage the NDR element actually sees is $m_i = v_i - s_i$, so

$$i_i = \phi_i(v_i - s_i,\, \xi_i), \qquad \dot{\xi}_i = q_i(v_i - s_i,\, \xi_i). \tag{9}$$

In vector form,

$$i = \phi(v - s,\, \xi), \qquad \dot{\xi} = q(v - s,\, \xi). \tag{10}$$

3.2 Kirchhoff's laws and the conductance Laplacian

Ohm's law on each resistive edge gives [2]

$$j = G\, B^T v. \tag{11}$$

Kirchhoff's current law says the net current flowing back into node i from the network is $-(Bj)_i$. Substituting Ohm's law,

$$i_{\text{net}} = -Bj = -BGB^T v = -L_G v, \tag{12}$$

where

$$L_G = BGB^T \in \mathbb{R}^{N\times N}. \tag{13}$$

This is the *weighted conductance Laplacian* of the intermediate network [16, 18]. Its emergence here is not a modelling choice; it is the unavoidable consequence of combining Ohm's law and KCL on any resistor network, whatever the topology. Concretely, the off-diagonal entry $(L_G)_{ij}$ for $i \neq j$ is $-g_{ij}$ (the conductance of the resistor between nodes i and j, or zero if they are not connected), and the diagonal entry $(L_G)_{ii} = \sum_{j\neq i} g_{ij}$ is the total conductance attached to node i. The matrix is symmetric and positive semi-definite; it is positive definite when the graph is connected; and its null space is spanned by the all-ones vector $\mathbf{1}$, expressing the obvious physical fact that a uniform voltage shift drives no current because all voltage differences vanish.

Applying KCL at node i gives

$$C_i \dot{v}_i = -i_i - (L_G v)_i. \tag{14}$$

Substituting (9) and stacking into vectors, the exact equations of motion are

$$C\dot{v} = -\phi(v - s, \xi) - L_G v, \tag{15}$$

$$\dot{\xi} = q(v - s, \xi), \tag{16}$$

with $C = \mathrm{diag}(C_1, \dots, C_N)$. These two equations, derived from nothing beyond Ohm's law and KCL, are the exact equations of motion for any network of NDR memristive elements connected by resistors.

Writing $L_G = \bar{g}\, L$ and assuming identical nodal capacitances $C_i = C_0$, the voltage equation becomes

$$\dot{v} = -\frac{1}{C_0}\phi(v - s, \xi) - \varepsilon L v, \qquad \varepsilon = \frac{\bar{g}}{C_0}. \tag{17}$$

The dimensionless coupling strength ε is simply the ratio of the inter-node conductance scale to the effective nodal capacitance.

4 The FitzHugh–Nagumo equations and their NDR realisation

4.1 The FitzHugh–Nagumo equations

The FitzHugh–Nagumo (FN) equations are the canonical two-dimensional reduction of the Hodgkin–Huxley model [3], introduced independently by FitzHugh [5] and by Nagumo, Arimoto, and Yoshizawa [6]. The key simplification is geometric: keep the fast voltage variable x with its folded-cubic nullcline and a slow recovery variable y with a linear nullcline, and everything else follows. The system is

$$\dot{x} = x - \frac{x^3}{3} - y + z, \qquad \dot{y} = \frac{1}{\tau}(x + a - by), \tag{18}$$

where z is the applied stimulus and τ, a, b are dimensionless parameters. The cubic term is what gives the voltage nullcline its N-shape, which is responsible for bistability and, when the fixed point lands on the unstable middle branch, for the limit-cycle oscillations that model repetitive spiking [8].

4.2 Identifying ϕ and ξ with FN variables

The I–V characteristic of the Pei et al. motif has exactly the folded-cubic form the FN model needs (Fig. 2).

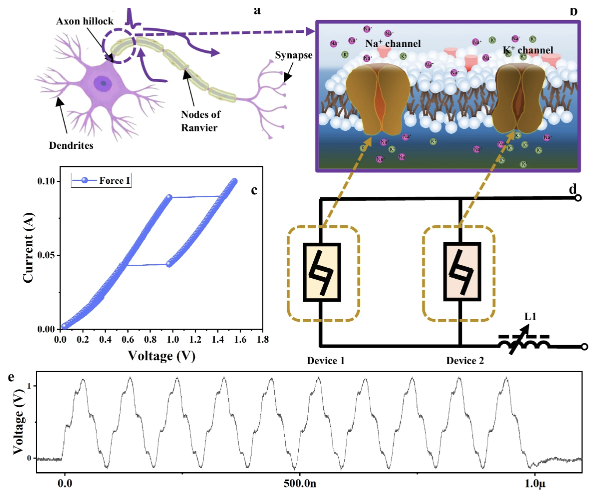

Figure 2: The I–V characteristic of the two-NDR-memristor motif (reproduced from [7]), showing the N-shaped negative differential resistance region and the box-type hysteresis loop.

Let $x(t)$ be the dimensionless voltage variable and $y(t)$ the dimensionless recovery variable. The internal state reduces to $\xi = y$, and the device functions take the specific forms

$$\phi(v-s,y) \;\longleftrightarrow\; -c\left(z+x-\frac{x^3}{3}-y\right), \qquad q(v-s,y) \;\longleftrightarrow\; \frac{1}{c}(x-a-by), \tag{19}$$

where $z = s/V_{\text{scale}}$ and $c > 0$ is the time-scale ratio. The minus sign in the first identification comes from the sign convention in (14): the NDR branch current *leaves* the node, so the current charging the capacitor is $-\phi_i$. With this identification, the isolated-motif equations become

$$\dot{x} = c\left(z+x-\frac{x^3}{3}-y\right), \tag{20}$$

$$\dot{y} = \frac{1}{c}(x-a-by). \tag{21}$$

We collect the right-hand side into the vector field

$$G(x,y) := \begin{pmatrix} c\left(z+x-\dfrac{x^3}{3}-y\right) \\ \dfrac{1}{c}(x-a-by) \end{pmatrix}, \tag{22}$$

so the isolated motif satisfies $(\dot{x},\dot{y})^T = G(x,y)$.

The time-scale separation is transparent here. The fast equation scales with c; the slow equation runs on time scale c, which is long when $c \gg 1$. In the Pei et al. circuit, this separation arises because D_1 switches fast (Na^+-like) and D_2 switches on a longer time scale set by its higher threshold voltage, with L_1 controlling the oscillation frequency [7].

4.3 The coupled FN network in dimensionless form

Substituting the identification (19) into (17), the coupled network reads

$$\dot{x}_i = c\left(z_i + x_i - \frac{x_i^3}{3} - y_i\right) - \varepsilon\,(Lx)_i, \tag{23}$$

$$\dot{y}_i = \frac{1}{c}(x_i - a - by_i). \tag{24}$$

In vector form,

$$\dot{x} = c\big(z + x - \tfrac{1}{3}x^{\circ 3} - y\big) - \varepsilon\,Lx, \tag{25}$$

$$\dot{y} = \frac{1}{c}(x - a\mathbf{1} - by). \tag{26}$$

Notice that coupling appears only in the voltage equation. This is physically correct: the resistors link the exposed voltage nodes and cannot directly drive the recovery variable y_i, which is internal to each motif. Setting $X_i = (x_i, y_i)^T$ and $X = (X_1, \ldots, X_N)$, the system takes the abstract form

$$\dot{X} = \mathcal{F}(X, \varepsilon) = F_0(X) + \varepsilon H(X), \tag{27}$$

where $F_0(X) = (G(X_1), \ldots, G(X_N))$ is the product of N uncoupled motif vector fields and $H(X)$ places $-(Lx)_i$ in the voltage slot of motif i and zero in the recovery slot.

4.4 Voltage sources in the intermediate network

A natural extension is to place a voltage source E_e in series with each coupling resistor R_e. From a neuroscience perspective, such sources model synaptic reversal potentials [4]: the voltage at which a synaptic current changes sign, set by the ionic composition of the junction. They can also model fixed electrochemical offsets or deliberate bias voltages introduced for circuit control.

With a source E_e on edge e, the effective voltage drop across that resistor is $(B^T v)_e - E_e$, so Ohm's law becomes $j = G(B^T v - e_{\mathrm{src}})$. The nodal injection current becomes

$$i_{\mathrm{net}} = -Bj = -L_G v + B_E, \tag{28}$$

where $B_E = BGe_{\mathrm{src}} \in \mathbb{R}^N$ is the *source-injection vector*. Note that $\mathbf{1}^T B_E = 0$: voltage sources between nodes inject no net current into the network, which is just Kirchhoff's current law at the network level. In dimensionless FN variables, B_E becomes an additive vector εu in the $\dot{x}$ equation:

$$\dot{x} = c\big(z + x - \tfrac{1}{3}x^{\circ 3} - y\big) - \varepsilon Lx + \varepsilon\,u, \tag{29}$$

$$\dot{y} = \frac{1}{c}(x - a\mathbf{1} - by), \tag{30}$$

and in abstract form,

$$\dot{X} = F_0(X) + \varepsilon H(X) + \varepsilon S(u), \tag{31}$$

where $S(u)$ has entries u_i in the voltage slots and zero in the recovery slots. Since $S(u)$ does not depend on X, the Jacobian $D_X\mathcal{F}$ is the same as in the source-free case. The sources shift the equilibria without touching their stability type.

Perturbative expansion for small sources

Write $u = \mu\hat{u}$ with $\|\hat{u}\| = 1$ and treat both ε and the source amplitude μ as small. At $(\varepsilon, \mu) = (0, 0)$ the equilibria are the uncoupled product points P_α defined below. Since $\mathcal{F}$ is C^1 and $D_X F_0(P_\alpha)$ is invertible, the Implicit Function Theorem applies jointly in (ε, μ) and gives a C^1 branch $P_\alpha(\varepsilon, \mu)$ with first-order expansion

$$P_\alpha(\varepsilon, \mu) = P_\alpha - \varepsilon[D_X F_0(P_\alpha)]^{-1} H(P_\alpha) - \varepsilon\mu[D_X F_0(P_\alpha)]^{-1} S(\hat{u}) + O(\varepsilon^2, \varepsilon\mu^2). \tag{32}$$

Several things are worth noting here. There is no $O(\mu)$ term at all: because u enters multiplied by ε, sources have no effect on equilibria when the resistors are disconnected, regardless of how large u is. This is obvious physically — voltage sources in an open circuit carry no current and do nothing. The leading effect of sources on equilibria is at order $\varepsilon\mu$. At this order the coupling correction and the source correction are additive. And because $S(u)$ is independent of X, the Jacobian at the perturbed equilibrium is the same as without sources: the equilibria shift but their stability type is unchanged.

The resolvent $[DG(p_j)]^{-1}$ has a closed form at equilibrium $p_j = (x_j, y_j)$:

$$[DG(p_j)]^{-1} = \frac{1}{1 - b + bx_j^2} \begin{pmatrix} -b/c & c \\ -1/c & c(1 - x_j^2) \end{pmatrix}. \tag{33}$$

The voltage and recovery shifts at node i due to a source E_e on edge $e = (i, j)$ are, to leading order in $\varepsilon\mu$,

$$\Delta x_{\alpha_i} \approx \frac{\varepsilon\mu\, g_e\, u_e\, b/c}{1 - b + bx_{\alpha_i}^2}, \qquad \Delta y_{\alpha_i} \approx \frac{\varepsilon\mu\, g_e\, u_e/c}{1 - b + bx_{\alpha_i}^2}, \tag{34}$$

both of which vanish as $\varepsilon \to 0$ or $\mu \to 0$, and both of which are well defined whenever $\det DG(p_{\alpha_i}) \neq 0$.

5 Equilibria and nullclines of the isolated motif

The equilibria of (20)–(21) are where both time derivatives vanish. Setting $\dot{x} = 0$ gives the *voltage nullcline*,

$$y_* = z + x_* - \frac{x_*^3}{3}, \tag{35}$$

and setting $\dot{y} = 0$ gives the *recovery nullcline*,

$$y_* = \frac{x_* - a}{b}. \tag{36}$$

Eliminating y_* gives the cubic

$$-\frac{x_*^3}{3} + \left(1 - \frac{1}{b}\right) x_* + \left(z + \frac{a}{b}\right) = 0. \tag{37}$$

Three distinct real roots — hence three equilibria $p_j = (x_j, y_j)$ with $x_1 < x_2 < x_3$ — occur when the slope $1/b$ of the recovery line falls between the local maximum and minimum slopes of the cubic and the stimulus z is in the bistable window.

The Jacobian at any point (x, y) is

$$DG(x, y) = \begin{pmatrix} c(1 - x^2) & -c \\ 1/c & -b/c \end{pmatrix}. \tag{38}$$

At equilibrium $p_j = (x_j, y_j)$,

$$\operatorname{tr} DG(p_j) = c(1 - x_j^2) - \frac{b}{c}, \tag{39}$$

$$\det DG(p_j) = 1 - b + bx_j^2. \tag{40}$$

The Routh–Hurwitz criterion for a 2×2 matrix reduces to two conditions: trace negative and determinant positive. So p_j is Hurwitz if and only if

$$c(1 - x_j^2) - \frac{b}{c} < 0 \quad \text{and} \quad 1 - b + bx_j^2 > 0. \tag{41}$$

The first condition is automatically satisfied whenever $|x_j| > 1$, i.e. whenever the equilibrium is on the outer branches of the cubic where the NDR is active. The middle equilibrium, when $M = 3$, always has $\det DG(p_2) < 0$: it is a saddle, always unstable.

6 Persistence and stability under weak coupling

Here is the analytical argument that the resting states survive coupling. We state it for a general smooth family of ODEs parameterised by a small coupling constant, because the same machinery will be reused twice — once for coupling alone and once for the joint coupling-plus-sources problem.

Consider

$$\dot{X} = \mathcal{F}(X, \varepsilon), \qquad X \in \mathbb{R}^n, \quad \varepsilon \in \mathbb{R}, \tag{42}$$

and suppose X^* is a resting state at $\varepsilon = 0$. The linearisation there is $\dot{\xi} = A\xi$ with $A = D_X\mathcal{F}(X^*, 0)$. We call X^* *Hurwitz* if all eigenvalues of A have strictly negative real part, and *hyperbolic* if no eigenvalue is purely imaginary.

Hurwitz implies hyperbolic, and it implies $\det A \neq 0$ because zero is on the imaginary axis and cannot be an eigenvalue of a Hurwitz matrix. This nonsingularity is the key: it is exactly the condition needed to apply the Implicit Function Theorem and show that the resting state moves continuously with ε rather than disappearing.

Theorem 1 (Persistence of hyperbolic equilibria)**.** *Let $\mathcal{F} : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ be C^1. Suppose $\mathcal{F}(X^*, 0) = 0$ and $\det D_X\mathcal{F}(X^*, 0) \neq 0$. Then there exist an open ball $U \ni X^*$ and $\varepsilon_0 > 0$ such that for each $|\varepsilon| < \varepsilon_0$ there is a unique $X^*(\varepsilon) \in U$ with $\mathcal{F}(X^*(\varepsilon), \varepsilon) = 0$. The map $\varepsilon \mapsto X^*(\varepsilon)$ is C^1 and $X^*(0) = X^*$.*

Proof. Apply the Implicit Function Theorem [13, 14] to $\mathcal{F}$ at $(X^*, 0)$. The hypothesis is exactly the invertibility of the partial Jacobian required by that theorem. □

This is the classical statement that hyperbolic fixed points are structurally stable [10, 11]: small smooth perturbations cannot destroy them. Knowing the continued fixed point $X^*(\varepsilon)$ exists is only half the story, though. We also need it to stay stable.

Theorem 2 (Persistence of Hurwitz stability)**.** *Under the hypotheses of Theorem 1, if additionally $D_X\mathcal{F}(X^*, 0)$ is Hurwitz, then there exists $0 < \varepsilon_1 \leq \varepsilon_0$ such that $X^*(\varepsilon)$ is Hurwitz and locally asymptotically stable for all $|\varepsilon| < \varepsilon_1$.*

Proof. Let $\gamma = -\max_\lambda \operatorname{Re}(\lambda)|_{\varepsilon=0} > 0$ be the spectral gap. The Jacobian $A(\varepsilon) = D_X\mathcal{F}(X^*(\varepsilon), \varepsilon)$ is continuous in ε because $\mathcal{F}$ is C^1 and $X^*(\varepsilon)$ is C^1. Eigenvalues are continuous functions of matrix entries — they are roots of the characteristic polynomial, and polynomial roots depend continuously on their coefficients [15]. So if all eigenvalues of $A(0)$ lie strictly in the open left half-plane, they cannot jump to the right as ε is turned on; they must stay there for some finite interval. Concretely, there exists ε_1 such that all eigenvalues of $A(\varepsilon)$ have real part at most $-\gamma/2 < 0$ for $|\varepsilon| < \varepsilon_1$. Local asymptotic stability follows from Lyapunov's indirect method [12] via local expansion near a fixed point. □

It is instructive to also have a proof that does not rely on the abstract continuity of eigenvalues, but instead makes the geometry more explicit via the Gershgorin discs.

Alternative proof via Gershgorin after Schur scaling. Since $A(0)$ is Hurwitz, every eigenvalue lies strictly in the open left half-plane. Let $\gamma > 0$ be the spectral gap from the imaginary axis. The Gershgorin discs of $A(0)$ need not lie in the left half-plane in the original coordinates, but a similarity transformation can fix that.

By the complex Schur decomposition, there exists a unitary Q such that $Q^*A(0)Q = T$ is upper triangular with diagonal entries $T_{ii} = \lambda_i$, the eigenvalues of $A(0)$, each satisfying $\operatorname{Re} T_{ii} \le -\gamma$. Introduce the diagonal scaling

$$D_\delta = \operatorname{diag}(1, \delta, \delta^2, \ldots, \delta^{n-1}), \qquad 0 < \delta < 1, \tag{43}$$

and define $T_\delta = D_\delta^{-1} T D_\delta$. This matrix is similar to $A(0)$ and has the same eigenvalues. Because T is upper triangular, the diagonal entries are unchanged: $(T_\delta)_{ii} = T_{ii} = \lambda_i$. For $j > i$, however, the off-diagonal entries transform as $(T_\delta)_{ij} = \delta^{j-i} T_{ij}$. By choosing δ small enough, we can make the row sums of off-diagonal entries uniformly smaller than $\gamma/2$:

$$\sum_{j \neq i} |(T_\delta)_{ij}| < \frac{\gamma}{2} \qquad \text{for all } i. \tag{44}$$

The i-th Gershgorin disc of T_δ is then centred at λ_i (real part $\le -\gamma$) with radius less than $\gamma/2$, so every point in every disc satisfies

$$\operatorname{Re} z \le -\gamma + \frac{\gamma}{2} = -\frac{\gamma}{2} < 0. \tag{45}$$

Now define $\widetilde{A}(\varepsilon) = D_\delta^{-1} Q^* A(\varepsilon) Q D_\delta$, which is similar to $A(\varepsilon)$ and hence has the same eigenvalues. Since $A(\varepsilon) \to A(0)$ as $\varepsilon \to 0$, we have $\widetilde{A}(\varepsilon) \to T_\delta$. Because the Gershgorin discs of T_δ lie a positive distance from the imaginary axis, those of $\widetilde{A}(\varepsilon)$ also remain in the open left half-plane for all sufficiently small $|\varepsilon|$. Therefore $A(\varepsilon)$ is Hurwitz for small ε, and local asymptotic stability follows from Lyapunov's indirect method. □

Applying these theorems to N uncoupled copies of the same system: the product system $\dot{X} = F_0(X)$ has exactly M^N equilibria, one for each way of assigning one of the M local resting states to each node. Label them

$$P_\alpha = (p_{\alpha_1}, \ldots, p_{\alpha_N}), \qquad \alpha \in \{1, \ldots, M\}^N. \tag{46}$$

The Jacobian at P_α is block diagonal,

$$DF_0(P_\alpha) = \operatorname{diag}(DG(p_{\alpha_1}), \ldots, DG(p_{\alpha_N})), \tag{47}$$

with determinant $\prod_i \det DG(p_{\alpha_i}) \neq 0$ and spectrum equal to the union of the individual spectra.

Corollary 1 (Product equilibria)**.** *If $G : \mathbb{R}^m \to \mathbb{R}^m$ is C^1 with M distinct hyperbolic equilibria, then $\dot{X} = F_0(X)$ has exactly M^N hyperbolic product equilibria. Every P_α whose component equilibria are all Hurwitz is itself Hurwitz.*

Applying Theorems 1 and 2 to the perturbed product system: all M^N product equilibria survive and those that were Hurwitz remain Hurwitz, for ε small enough.

7 Equilibria and stability of the coupled network

We now apply the abstract results directly to the NDR–FN network (27).

Assumption 1. *The FN parameters (a, b, c, z) are such that the isolated motif has M distinct equilibria $p_1, \ldots, p_M$, each hyperbolic. In practice, either the motif is monostable ($M = 1$, one resting state) or bistable ($M = 3$, two stable outer resting states separated by a saddle).*

Under this assumption, Corollary 1 gives M^N hyperbolic product equilibria for the uncoupled network. Their Jacobian determinant factorises as

$$\det D_X F_0(P_\alpha) = \prod_{i=1}^{N}(1 - b + bx_{\alpha_i}^2) \neq 0, \tag{48}$$

each factor being $\det DG(p_{\alpha_i})$, nonzero by hyperbolicity.

Theorem 3 (Persistence and stability for the NDR–FN network)**.** *Let Assumption 1 hold. Consider the Laplacian-coupled network* (25)–(26) *with coupling strength $\varepsilon = \bar{g}/C_0$. For each multi-index $\alpha = (\alpha_1, \ldots, \alpha_N) \in \{1, \ldots, M\}^N$, let $P_\alpha = (p_{\alpha_1}, \ldots, p_{\alpha_N})$ be the corresponding product equilibrium of the uncoupled network. Then:*

(i) (Persistence.) *For each α, there exist $\varepsilon_\alpha > 0$ and a neighbourhood U_α of P_α such that, for every $|\varepsilon| < \varepsilon_\alpha$, the coupled network has a unique equilibrium $P_\alpha(\varepsilon) \in U_\alpha$ with $P_\alpha(0) = P_\alpha$, depending C^1-smoothly on ε.*

(ii) (Stability.) *If $p_{\alpha_1}, \ldots, p_{\alpha_N}$ are all Hurwitz, then there exists $0 < \varepsilon_\alpha^{\mathrm{H}} \leq \varepsilon_\alpha$ such that $P_\alpha(\varepsilon)$ is Hurwitz and locally asymptotically stable for all $|\varepsilon| < \varepsilon_\alpha^{\mathrm{H}}$.*

(iii) (Uniform threshold.) *If every p_j is Hurwitz, both thresholds may be chosen uniformly in α: there exists $\varepsilon_* > 0$ such that for $|\varepsilon| < \varepsilon_*$ all M^N product equilibria continue to distinct nearby Hurwitz equilibria of the coupled network.*

Proof. Write $\mathcal{F}(X, \varepsilon) = F_0(X) + \varepsilon H(X)$. At $\varepsilon = 0$ the motifs are uncoupled, so $\mathcal{F}(P_\alpha, 0) = 0$ for every α.

Persistence. The partial Jacobian at $(P_\alpha, 0)$ is $D_X\mathcal{F}(P_\alpha, 0) = D_X F_0(P_\alpha) = \mathrm{diag}(DF(p_{\alpha_1}), \ldots, DF(p_{\alpha_N}))$. Each block is invertible by Assumption 1, so the determinant of the whole matrix is nonzero by (48). The Implicit Function Theorem then provides the neighbourhood U_α, the radius $\varepsilon_\alpha > 0$, and the unique C^1 branch $P_\alpha(\varepsilon)$. This proves (i).

Stability. If $p_{\alpha_1}, \ldots, p_{\alpha_N}$ are Hurwitz, the Jacobian $A_\alpha(0) = \mathrm{diag}(DF(p_{\alpha_1}), \ldots, DF(p_{\alpha_N}))$ is Hurwitz with spectral gap $\gamma_\alpha > 0$. Since $\mathcal{F}$ is C^1 and $P_\alpha(\varepsilon) \to P_\alpha$, the Jacobian $A_\alpha(\varepsilon) = D_X\mathcal{F}(P_\alpha(\varepsilon), \varepsilon)$ is continuous in ε. Continuity of eigenvalues gives $\varepsilon_\alpha^{\mathrm{H}} > 0$ such that all eigenvalues of $A_\alpha(\varepsilon)$ have real part at most $-\gamma_\alpha/2 < 0$ for $|\varepsilon| < \varepsilon_\alpha^{H}$. Lyapunov's indirect method gives local asymptotic stability. This proves (ii).

Uniform threshold. Take $\varepsilon_* = \min_\alpha \varepsilon_\alpha^{\mathrm{H}}$, a minimum over a finite set, hence positive. The neighbourhoods U_α may be chosen pairwise disjoint; for $|\varepsilon| < \varepsilon_*$ the continued branches remain inside these disjoint sets and are therefore distinct. This proves (iii). □

A comment on what controls ε_* is worth making. Its value is governed by two physical quantities: the spectral gap $\gamma = \min_\alpha \gamma_\alpha$, which measures how far the resting states are from bifurcation, and the operator norm $\|L\|$, which measures how strongly the coupling Laplacian acts. A larger spectral gap and a smaller Laplacian norm both allow a larger coupling before stability is lost. Explicit, if conservative, lower bounds on ε_* can be extracted from matrix perturbation theory [15].

The theorem is local: it guarantees one continued equilibrium near each uncoupled product state, but says nothing about what happens far away in phase space. A global statement — that the coupled system has *exactly* M^N equilibria — would require additional control on the vector field away from the product equilibria.

In the bistable case $M = 3$, the theorem delivers 2^N stable product equilibria, one for each binary assignment of "resting" or "excited" to the N nodes. This 2^N scaling is the direct physical consequence of the bistability of each NDR motif: each node acts as a one-bit memory element, and weak coupling does not disrupt this.

To briefly recap where we are: we started at the device level with the NDR memristive element, assembled it into the FitzHugh–Nagumo motif, and derived the network equations of motion from Kirchhoff's laws alone. The result is a smooth $O(\varepsilon)$ deformation of the direct product of N uncoupled FN systems, with the deformation controlled by the conductance Laplacian $L_G = BGB^T$ and the dimensionless coupling strength $\varepsilon = \bar{g}/C_0$. Two applications of standard tools — the IFT for existence and eigenvalue continuity for stability — then show that the M^N resting states all survive and remain stable for $\varepsilon < \varepsilon_*$.

The voltage-source extension fits naturally into the same framework: sources appear as additive forcing εu in the voltage equation, shift the resting states at order $\varepsilon\mu$ as given explicitly by (32)–(34), and leave stability type unchanged.

8 Connection with neural ODEs

The coupled NDR–FN network also fits directly into the neural ODE framework [21]. In a neural ODE, one specifies a parameterised vector field $\dot{X} = f_\theta(X)$ and computation is performed by the flow map $X(T) = \Phi_T^\theta(X(0))$; training adjusts θ so that $X(T)$ has the desired behaviour. Our network is exactly that, but with a physically motivated constraint on how θ enters. The state is $X = (x, y) \in \mathbb{R}^{2N}$, and the parameterised dynamics are

$$\dot{x} = c\left(z + x - \frac{1}{3}x^{\circ 3} - y\right) - \varepsilon L_\theta x + \varepsilon u, \tag{49}$$

$$\dot{y} = \frac{1}{c}(x - a\mathbf{1} - by), \tag{50}$$

with

$$L_\theta = B\Gamma B^T, \qquad \Gamma = \mathrm{diag}(g_1, \ldots, g_E), \qquad g_e \geq 0, \tag{51}$$

so the trainable parameters are the edge conductances g_e and the optional bias vector u:

$$\theta = (g_1, \ldots, g_E, u). \tag{52}$$

This is a *physics-constrained neural ODE*. The local nonlinear part of the vector field is fixed by the NDR–FN circuit; only the coupling is trainable, and it is constrained to be a passive Laplacian:

$$g_e \geq 0, \quad L_\theta = L_\theta^T, \quad L_\theta \mathbf{1} = 0, \quad (L_\theta)_{ij} \leq 0 \quad i \neq j. \tag{53}$$

These constraints encode the fact that the intermediate network is made of ordinary resistors.

Memory recall in this picture is a computation performed by the flow. A corrupted input pattern initialises the state, $X(0) = X_{\text{corr}}$, and the trained dynamics carry it to $X(T) = \Phi_T^\theta(X_{\text{corr}})$.

The voltage component $x(T)$ is then thresholded to obtain the recalled bit pattern. The system computes not by a feedforward sequence of matrix multiplications but by physical relaxation in continuous time.

This viewpoint also clarifies the role of the training objectives in the next section. The fixed-point loss trains the vector field so that the desired memory patterns are equilibria. The stability loss trains those equilibria to be attracting. The trajectory or basin loss trains the finite-time flow map Φ_T^θ so that corrupted inputs converge to the intended memories. The learned resistor mesh plays the role of the trainable weights of a recurrent neural ODE, while the NDR–FN motifs provide the nonlinear activation and local bistability.

9 Training the resistive mesh as an associative memory

The trainable parameters are the conductances of the intermediate resistive mesh, together with optional source injections. The objective is to shape the vector field so that the desired patterns become attracting equilibria of the continuous-time dynamics. The NDR–FN motifs provide the local nonlinear state space and bistability; the resistor mesh provides the learnable interaction structure. A stored memory is an attracting fixed point of the ODE. Recall is the relaxation of the flow toward that fixed point from a corrupted initial condition, as in continuous-time associative memories [22, 23].

We use the source-augmented network (29)–(30).

9.1 Patterns as target equilibria

Let $\beta^1, \ldots, \beta^P \in \{0,1\}^N$ be the binary patterns to be stored. Choose a voltage amplitude $r > 0$ and encode each pattern as

$$\mu^p = r(2\beta^p - \mathbf{1}) \in \mathbb{R}^N, \tag{54}$$

so bit 0 maps to voltage $-r$ and bit 1 to voltage $+r$. The value of r is a modelling choice. Setting r equal to the natural stable voltage $x_\pm$ of the isolated motif is convenient because $h(\pm r) = 0$ exactly, which removes a systematic term from the training problem. But it is sometimes useful to choose r slightly off the natural equilibrium, so the local NDR–FN force is not identically zero at the target and the mesh has a nontrivial force to balance.

To make μ^p a fixed point of the voltage dynamics, the recovery variable is pinned by (30) at

$$\nu^p = \frac{\mu^p - a\mathbf{1}}{b}. \tag{55}$$

Substituting $x = \mu^p$ and $y = \nu^p$ into (29) gives the fixed-point condition

$$h(\mu^p) - \varepsilon L_\theta \mu^p + \varepsilon u = 0, \tag{56}$$

where

$$h(\mu) := c\left(z + \mu - \frac{1}{3}\mu^{\circ 3} - \frac{\mu - a\mathbf{1}}{b}\right) \tag{57}$$

is the residual force from the local NDR–FN motifs at voltage pattern μ, after the recovery variable has been eliminated. Using the edge expansion $L_\theta = \sum_e g_e b_e b_e^T$, the fixed-point condition becomes

$$\varepsilon \sum_{e=1}^{E} g_e b_e b_e^T \mu^p - \varepsilon u = h(\mu^p), \qquad p = 1, \ldots, P. \tag{58}$$

This is the algebraic core of training: although the ODE is nonlinear in the state variables, the exact fixed-point constraints are linear in the conductances g_e and in u.

9.2 Algebraic pretraining by nonnegative least squares

Collecting the constraints into matrices gives

$$\varepsilon L_\theta M - \varepsilon u \mathbf{1}_P^T = H, \tag{59}$$

where $M = [\mu^1 \cdots \mu^P]$ and $H = [h(\mu^1) \cdots h(\mu^P)]$. A natural first stage of training is to solve the convex problem

$$\min_{g_e \geq 0,\, u} \left\| \varepsilon \sum_e g_e b_e b_e^T M - \varepsilon u \mathbf{1}_P^T - H \right\|_F^2 + \lambda_g \sum_e g_e^2 + \lambda_u \|u\|^2. \tag{60}$$

The first term is the fixed-point loss: it measures how far the desired patterns are from being exact equilibria of the trained neural ODE.

Algorithm 1. Passive fixed-point training and recall test

Input: binary patterns $\beta^1, \ldots, \beta^P \in \{0,1\}^N$, graph template $\mathcal{G} = (V, E)$, FN parameters (a, b, c, z), coupling strength ε, voltage amplitude r, regularisation parameters λ_g, λ_u, and recall corruption probability ρ.

Output: learned conductances $g_e \geq 0$, bias vector u, refined fixed points Q^p, stability diagnostics, and recall errors.

1. Encode each binary pattern as a voltage memory $\mu^p = r(2\beta^p - \mathbf{1})$ and set the target recovery coordinate $\nu^p = (\mu^p - a\mathbf{1})/b$.

2. Choose an orientation for the graph and form its incidence vectors b_e. The trainable Laplacian is $L_\theta = \sum_{e \in E} g_e b_e b_e^T$, with the physical constraint $g_e \geq 0$.

3. Evaluate the local force $h(\mu^p) = c(z + \mu^p - (\mu^p)^{\circ 3}/3 - \nu^p)$ for every target pattern. This is the force that the passive mesh and bias must cancel at the desired equilibrium.

4. Stack the linear constraints

$$\varepsilon \sum_e g_e b_e b_e^T \mu^p - \varepsilon u = h(\mu^p), \qquad p = 1, \ldots, P,$$

into a least-squares system for the unknowns (g, u). Solve the nonnegative least-squares problem (60), constraining only the conductances to be nonnegative while leaving the node bias u unconstrained.

5. Assemble the trained network using the learned L_θ and u. For each memory, initialise Newton's method at (μ^p, ν^p) and refine it to a true fixed point $Q^p = (x^p, y^p)$ of the full $2N$-dimensional ODE.

6. Evaluate the residual $\|F_\theta(Q^p)\|$ and the eigenvalues of the Jacobian $DF_\theta(Q^p)$. The memory is accepted as a stable learned attractor when the residual is below tolerance and

$$\max_{\lambda \in \sigma(DF_\theta(Q^p))} \operatorname{Re} \lambda < 0.$$

7. For recall, corrupt each stored bit pattern by independent bit flips with probability ρ, initialise the ODE from the corrupted voltage pattern, integrate to time T, and threshold the final voltage by sign to obtain $\widehat{\beta}$.

8. Report the learned mesh, fixed-point residuals, stability spectra, exact-recall indicators, and Hamming errors $d_{\mathrm{H}}(\widehat{\beta}, \beta^p)$.

Proposition 1 (Algebraic fixed-point training)**.** *Let $\mu^1, \ldots, \mu^P \in \mathbb{R}^N$ be target voltage patterns. Suppose there exist conductances $g_e \geq 0$ and a vector u such that* (58) *holds for every $p = 1, \ldots, P$. Let $L_\theta = \sum_e g_e b_e b_e^T$. Then each point*

$$Q^p = \left(\mu^p, \frac{\mu^p - a\mathbf{1}}{b}\right) \in \mathbb{R}^{2N} \tag{61}$$

is an equilibrium of (29)–(30)*.*

Proof. Fix p and set $x = \mu^p$, $y = (\mu^p - a\mathbf{1})/b$. Then $\dot{y} = 0$ by construction. The voltage equation gives $\dot{x} = h(\mu^p) - \varepsilon L_\theta \mu^p + \varepsilon u$, which vanishes by (58). □

9.3 Compatibility restrictions imposed by passivity

The passivity of L_θ imposes a non-obvious constraint on which patterns can be exactly stored. Since $L_\theta \mathbf{1} = 0$, multiplying (56) by $\mathbf{1}^T$ gives

$$\mathbf{1}^T h(\mu^p) + \varepsilon \mathbf{1}^T u = 0, \qquad p = 1, \ldots, P. \tag{62}$$

If $u = 0$, every exactly stored pattern must satisfy $\mathbf{1}^T h(\mu^p) = 0$. If a single shared bias u is used for all patterns, then all target patterns must have the same total local force:

$$\mathbf{1}^T h(\mu^1) = \cdots = \mathbf{1}^T h(\mu^P). \tag{63}$$

This is a necessary compatibility condition. If it fails, no passive mesh with a single shared u can store all targets as exact fixed points. At the natural amplitude $r = x_\pm$, where $h(\pm r) = 0$, the condition reduces to a simple bit-balance requirement: each pattern must have as many $+r$ nodes as $-r$ nodes (or, more generally, the same imbalance across all patterns if $u \neq 0$).

More generally, the set of learnable right-hand sides is the cone

$$\mathcal{C}_M = \left\{ \sum_{e=1}^{E} g_e b_e b_e^T M : g_e \geq 0 \right\}, \tag{64}$$

and exact learnability requires $(H + \varepsilon u \mathbf{1}_P^T)/\varepsilon \in \mathcal{C}_M$.

9.4 Stability of learned memories

A learned equilibrium is useful as a memory only if it is attracting. At the target state Q^p, the Jacobian of (29)–(30) is

$$J_p(\theta) = \begin{pmatrix} c\,\mathrm{diag}\big(1 - (\mu^p)^{\circ 2}\big) - \varepsilon L_\theta & -cI \\ \frac{1}{c} I & -\frac{b}{c} I \end{pmatrix}. \tag{65}$$

The memory is locally asymptotically stable if $J_p(\theta)$ is Hurwitz. For a desired margin $\rho > 0$, a stability regulariser is

$$\mathcal{L}_{\mathrm{stab}}(\theta) = \sum_{p=1}^{P} \left[\max\left\{ 0,\ \rho + \max_{\lambda \in \sigma(J_p(\theta))} \mathrm{Re}\,\lambda \right\} \right]^2. \tag{66}$$

Theorem 4 (Local associative-memory property)**.** *Suppose the learned parameters $\theta = (g, u)$ satisfy the exact fixed-point constraints* (58) *for patterns $\mu^1, \dots, \mu^P$. If each Jacobian $J_p(\theta)$ is Hurwitz, then every target state Q^p is a locally asymptotically stable memory of the trained neural ODE.*

Proof. Proposition 1 shows that each Q^p is an equilibrium. Since the Jacobian at Q^p is Hurwitz by assumption, Lyapunov's indirect method [12] implies local asymptotic stability. □

9.5 Trajectory-level recall training

The fixed-point loss controls where the equilibria are and the Hurwitz condition controls whether they are locally attracting, but associative memory also depends on basin geometry. A corrupted input should flow back to the intended memory. Let Φ_T^θ be the time-T flow map of the trained ODE. For a corrupted version $\tilde{\beta}^{p,r}$ of target bit pattern β^p, define

$$x^{p,r}(0) = r(2\tilde{\beta}^{p,r} - \mathbf{1}), \qquad y^{p,r}(0) = \frac{x^{p,r}(0) - a\mathbf{1}}{b}. \tag{67}$$

A direct trajectory-level recall loss is

$$\mathcal{L}_{\text{recall}}(\theta) = \sum_{p,r} \|x^{p,r}(T;\theta) - \mu^p\|^2. \tag{68}$$

This term trains the finite-time flow itself. It is generally nonconvex, because it depends on the nonlinear ODE solution, but it directly measures whether corrupted inputs return to the intended memories. The full training objective is

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{fp}}(\theta) + \lambda_{\text{stab}}\mathcal{L}_{\text{stab}}(\theta) + \lambda_{\text{recall}}\mathcal{L}_{\text{recall}}(\theta) + \lambda_g \sum_e g_e^2 + \lambda_u \|u\|^2, \tag{69}$$

subject to $g_e \geq 0$. In the simulations, the convex fixed-point problem is used as an algebraic pretraining stage and recall is then tested by integrating the ODE from corrupted initial conditions.

The recalled bit pattern is obtained by thresholding the final voltage:

$$\widehat{\beta}_i^{p,r} = \begin{cases} 1, & x_i^{p,r}(T;\theta) > 0, \\ 0, & x_i^{p,r}(T;\theta) \leq 0. \end{cases} \tag{70}$$

Exact recall occurs when $\widehat{\beta}^{p,r} = \beta^p$, equivalently when the Hamming error $d_{\text{H}}(\widehat{\beta}^{p,r}, \beta^p)$ vanishes.

Proposition 2 (Recall from a learned basin)**.** *Suppose Q^p is a locally asymptotically stable learned memory and X_0 lies in its basin of attraction. Then $\lim_{t\to\infty} x(t) = \mu^p$. If every component of μ^p is bounded away from zero, the thresholded state equals β^p for all sufficiently large T.*

Proof. Since X_0 lies in the basin of attraction of Q^p, the flow converges to Q^p and its voltage component converges to μ^p. Because each component of μ^p has nonzero sign margin, the sign of $x_i(t)$ agrees with the sign of μ_i^p for all sufficiently large t, and thresholding recovers β^p. □

9.6 What the architecture can and cannot store

The construction is Hopfield-like in that memories are stable attractors of a recurrent continuous-time system [22]. The difference from a generic neural ODE is that the learned coupling is constrained to be a passive Laplacian, not an arbitrary weight matrix. This means the patterns most naturally stored by this architecture are *graph-compatible* memories: consensus states, domains, smooth images on the mesh, community patterns, silhouettes — configurations whose restoring forces can be represented by nonnegative edge-difference forces. More arbitrary unstructured patterns may require additional degrees of freedom: trainable source injections u, hidden mesh nodes, active coupling elements, or an encoder–decoder architecture where the NDR–FN network stores latent patterns rather than raw images.

10 Numerical experiment: 8 by 8 digit memories

Here we give a first numerical test of the learning construction. The goal is not to compete with standard image-classification methods, but to check whether the resistive NDR–FN network can be trained so that a collection of binary images becomes a collection of locally stable fixed points, and whether noisy initial conditions flow back toward those fixed points under the physical dynamics. The training and validation pipeline follows Algorithm 1.

10.1 Experimental setup

We used an 8×8 voltage grid, hence $N = 64$ NDR–FN motifs, one per pixel. For the default parameters $a = 0$, $b = 2$, $c = 5$, $z = 0$, the natural stable voltages are $x_\pm = \pm\sqrt{3/2} \simeq \pm 1.2247$. Using this amplitude is convenient because $h(x_\pm) = 0$ exactly.

The training set was ten binary 8×8 digit prototypes, one for each digit $0, \ldots, 9$. In the run reported here the built-in synthetic digit templates were used; the same script can use the scikit-learn 8×8 handwritten digit dataset by selecting `--digit-dataset sklearn`. The graph template was complete, so the trainable mesh had $E = 64 \cdot 63/2 = 2016$ candidate conductances. Training solved the nonnegative least-squares problem for $g_e \geq 0$ and a trainable node bias u, so that each target image μ^p approximately satisfied

$$h(\mu^p) - \varepsilon L_\theta \mu^p + \varepsilon u = 0. \tag{71}$$

After training, each candidate memory was Newton-refined to the nearest true fixed point of the full $2N$-dimensional ODE, and the Jacobian was evaluated at that refined fixed point.

Figure 3: The ten 8×8 digit memories used as target fixed points. Each pixel is one NDR–FN memory element; white/blue pixels correspond to the two stable voltage states of the isolated motif.

10.2 Learned resistor mesh

The least-squares solve produced a sparse-ish passive mesh: 1234 of the 2016 candidate edges had conductance larger than 10^{-10}, with total conductance $\sum_e g_e = 33.07$. The residual at the intended target voltages was small,

$$\max_p \|h(\mu^p) - \varepsilon L_\theta \mu^p + \varepsilon u\| = 1.90 \times 10^{-8},$$

and the residual after Newton refinement to the actual network fixed points was at machine precision, $\max_p \|F(Q^p)\| = 3.03 \times 10^{-15}$. All ten refined fixed points were Hurwitz. The least stable one had $\max_\lambda \operatorname{Re}\lambda = -0.796$, so the learned memories are not merely algebraic solutions of the fixed-point equations: in this run they are linearly stable attractors of the trained NDR–FN network.

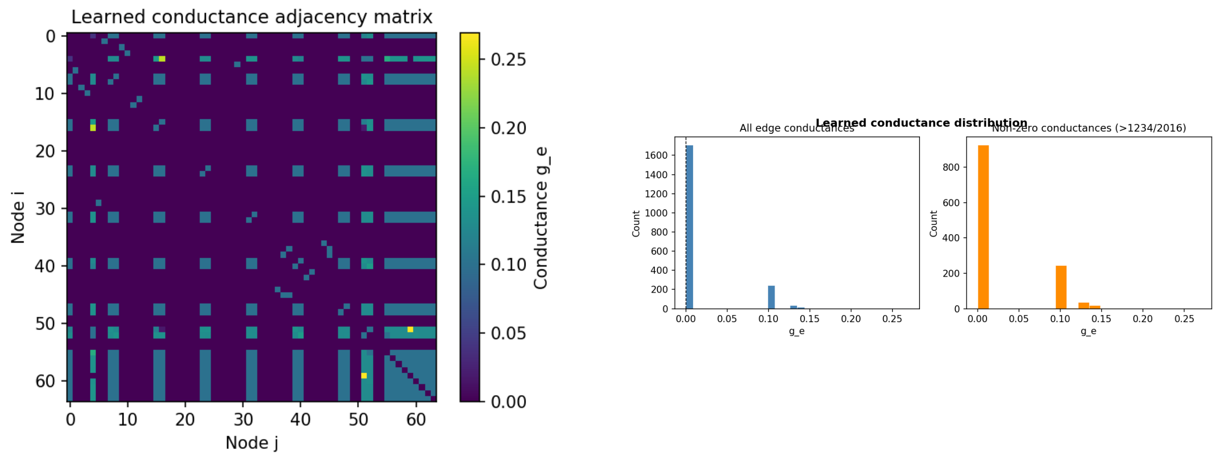

Figure 4: Left: learned conductance adjacency matrix of the complete 64-node template. Right: distribution of learned edge conductances. The passive constraint $g_e \geq 0$ is enforced during training.

10.3 Recall from corrupted initial conditions

To test recall, each target digit was corrupted by independently flipping bits with probability 0.05. The corrupted image was used as the initial condition, the trained network was integrated, and the final voltage was thresholded at zero. The mean input corruption was 2.6 pixels per digit. After relaxation, the mean final Hamming error was 1.9 pixels. Three of the ten trials were recalled exactly.

Digit	input errors	final errors	exact recall	stable final state
0	1	0	yes	yes
1	3	2	no	yes
2	3	3	no	yes
3	5	3	no	yes
4	1	0	yes	yes
5	4	3	no	yes
6	0	0	yes	yes
7	4	3	no	yes
8	3	3	no	yes
9	2	2	no	yes

Every recalled endpoint was a stable fixed point of the trained network, with residuals of order 10^{-15}. The failures are not integration failures: they are genuine basin effects, where the corrupted input was close enough to a different stable configuration that the flow ended there rather than at the intended digit. This is the standard spurious-attractor problem of Hopfield-type networks.

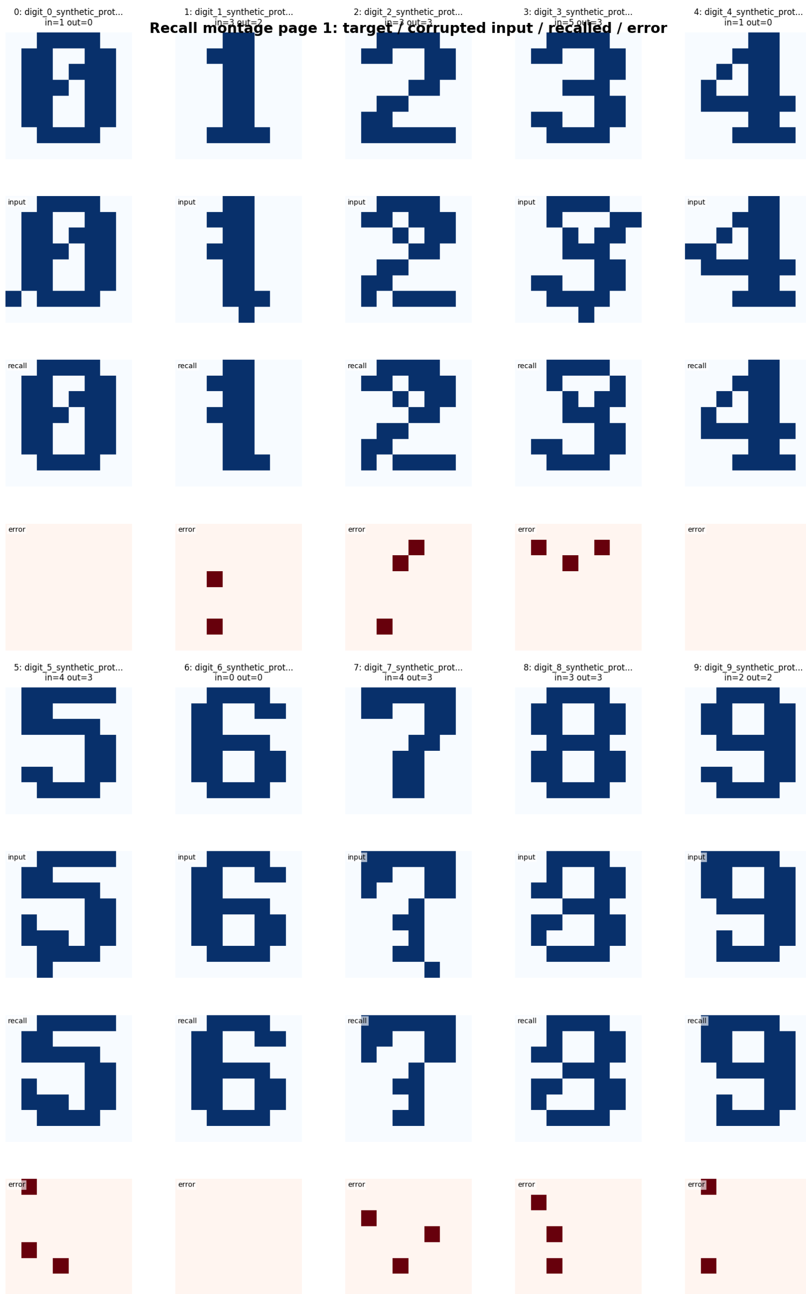

Figure 5: Recall montage for the ten digit memories. For each digit the panels show, from top to bottom, the stored target, the corrupted input used to initialise the ODE, the recalled output after relaxation, and the pixelwise error image $|\widehat{\beta} - \beta|$.

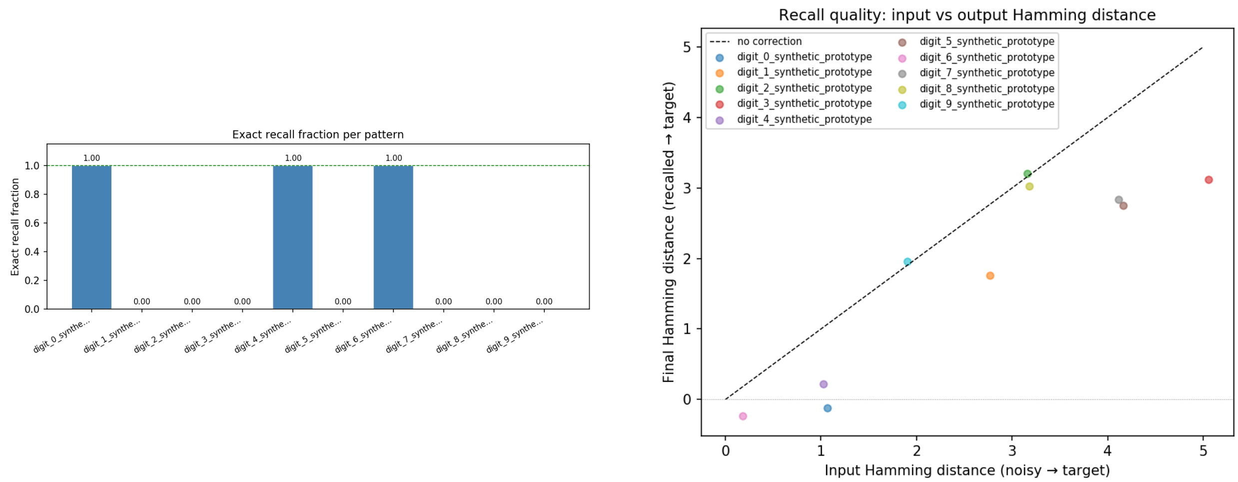

Figure 6: Recall diagnostics. Left: exact-recall fraction for each stored digit. Right: input Hamming corruption versus final Hamming error. Points below the diagonal indicate denoising by the NDR–FN flow.

10.4 What the experiment tells us

The experiment demonstrates the basic mechanism promised by the preceding sections. A passive resistor mesh can be trained so that a set of binary image patterns becomes a set of stable fixed points of the coupled NDR–FN network, and the fixed-point equations are solved at the level of the physical parameters — the conductances — not by any kind of black-box optimisation. The resulting memories remain stable under the full nonlinear dynamics.

The recall results also make the limitation concrete. Storing ten digit prototypes is straightforward for the complete passive mesh with bias: the fixed-point residuals are tiny and all Jacobians are Hurwitz. Perfect recall from corrupted images is harder, because it depends on basin geometry rather than just the existence and stability of the target equilibria. The natural next step, in the language of neural ODEs, is trajectory-level training of the flow map: include noisy versions of each target pattern in the loss, or add hidden nodes, structured graph templates, or active coupling elements to widen the basins of attraction.

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